Lab III

Relational Operators, Logical Operators and Conditionals

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>. Section B

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Learning objectives:

The following types will be covered in this lab session

- Comparison/Relational Operators
- Logical Operators
- If Statement
- If - else statement
- else - if Statement

Theory: Comparison/Relational Operators

Operator Name	Syntax
Less than	a<b
Less than equal to	a<=b
Greater than	a>b
Greater than equal to	a>=b
Equal to	a==b
Not equal to	a !=b

Logical operators:

Operator Name	Syntax
Logic Negation (not)	!a
Logic And	a && B
Logic or	a || b

Decision Control Structures:

- Till now we have used sequence control structure in the programs, in which the various steps are executed sequentially i.e. in the same order in which they appear in the program.
- In C++ programming the instructions are executed sequentially, by default. At times, we need a set of instructions to be executed Depending upon different conditions to be satisfied.
- In such cases we have to use decision control instructions. This can be achieved in C using;

1. The if statement
2. The if-else statement
3. The else - if statement ➪ **The C/C++ "if" Statement**

- The **if** statement controls conditional branching. The body of an if statement is executed if the value of the expression/condition specified in the if statement is true. The syntax of the if statement is as follows:

Syntax:

```
If (expression)
{
Block of statement;
}
```

➪ Example:

- Write a program in which it takes a number from keyboard as an input and if the number is greater than 100 it prints "The number is greater than hundred".

```
#include <iostream>
#include<conio.h>

int main()
{
    int number ;
    cout<< "Enter an integer\n";
    cin>> number;
    if ( number >100 )
        cout<<"The number is greater than 100";
    getch();

}
```

- Note we did not use curly brackets '{}'for body of if **if ()** because it did not include multiple statements (block of statements). If it includes multiple statements then it would have been something like this.

```
if (number<100){
    cout<< "The number is greater than 100\n";

 cout<< "No doubt that the number is greater than 10

    }
```

The C++ if-else Statement:

- **if - else** statement is similar to **if** with the addition of else statement. If the condition is false in **if** then its body will be skipped and the else statement's body will be executed.

Syntax:

```
if (expression)
{
  Block of statement;
}
else
{
  Block of statement;
}
```

Example:

- Write a program in which it takes two numbers from keyboard as input and subtract larger number from smaller.

```
#include <iostream>
#include<conio.h>


int main()
{
        int a,b ;

        cout << “Enter first number\n”;
        cin >> a;
        cout <<  “Enter second number\n”;
        cin >> b;

        if ( a >=b )
        // this condition can also be written as if(a>b || a==b)
                cout <<a <<"-" <<b <<"=" <<a-b;
        else
                cout <<b <<"-" <<a << "=" <<b-a;

        getch();

}
```

Nested if-else statement:

- Nested **if** is basically **if** inside **if**.

Example:

- Write a program which take a number from keyboard and checks the number whether that number is less than 100 or not if that number is less than 100 than check that is it less than 50 or not.

```
#include <iostream>
#include <conio.h>

int main()
{
        int number ;
        cout << "Enter an integer\n";
        cin >>number;
        if ( number <100 )
        {
                cout<< "Yes the number is less than 100";
                if ( number <50)
                {
                        cout<< " and number is also less than 50";
                }
                else
                {
                        cout<< " but the number is not less than 50";
                }
        }
        else
                cout<< "No the number is not less than 100";
        getch();

}
```

- The above program checks if the number is less than 100. This check is done in the first **if** statement.
- If the result is true the program enters the second **if** statement which is also called the **nested if** statement. Here another check is performed. The number is checked if it less than 50 or not. If the number is less than 50 it is conveyed to the user. The rest is pretty self-explanatory.

The C++ else-if Statement:

- Sometimes we wish to make a multi-way decision based on several conditions. The most general way of doing this is by using the else if variant on the if statement. This works by cascading several comparisons. As soon as one of these gives a true result, the following statement or block is executed, and no further comparisons are performed.

```
Syntax:
if(expression)
{
Block of statement;
} else
if(exprerssi
on)
{
Block of statement;
} else
if(exprerssi
on)
{
Block of statement;
}
. . // you can add as many else-if statements as many you need
. .
. .
else
{
Block of statement;
}
```

Lab Tasks:

- **Task 1: Comparison/Relational Operators:**

1. Write a C program that prompts the user to enter two numbers.
2. Use comparison operators to check if the first number is equal to, greater than, or less than the second number.
3. Display appropriate messages based on the comparison results.

Task 1: Comparison / Relational Operators

Program

```
#include <iostream>                      // tells the cmpiler to include
standard input output library
using namespace std;          // allows us to use cout and cin without
tyiping std:: for every one

int main(){                              //starting point of the main function
      int a,b;                     // variable declaration
      cout<<"enter an integer ";   // prompt for user to input data
      cin>>a;                          // takes input from user and places
it in the variable
      cout<<"enter 2nd integer ";        // 2nd prompt for user to input
data
      cin>>b;                          // takes data from user and places
it in the variable

      if(a>=b)                           // condition for true

    cout<<a<<"is greater than or equal to  "<<b; //  prints the result if the
if condition met
    else                                          // part which runs if
the if condition fails
    cout<<a<<"is less than "<<b;                //prints result for a
smaller than b

    return 0;                        // tells the operating system that the
program finished successfully


}
```

Explanation

This program asks the user to enter two numbers.
The relational operators **==, >, and** < are used to compare the numbers.

Depending on the comparison result, the program displays an appropriate message indicating whether the numbers are **equal, greater, or smaller**.

Task 2: Logical Operators

Program

```
#include <iostream>             // tells the compiler to include standard
input output library
using namespace std;               // allows us to use cout without having to
type std:: for each

int main(){                     // the starting point from where the computer
begins executing program

      int a;                       // variable declaration

      cout<<"enter an integer ";     // prompt for user to input data
      cin>>a;                           // takes data from user and puts it
into the variable

      if(a>0 && a<100)                  // condition for the program to
cheak a number is greater than 0 and less than 100
      cout<<"condition met between 0 and 100"<<endl;  // if the condition met
the message is sent

      else                                              // part that runs if
the if condition fails
      cout<<"condition not met"<<endl;        // if the condion doesnt met
this message is sent

      return 0;   // tells the operating system that program has finished
}
```

Explanation

This program checks whether the entered number is **greater than 0 and less than 100**.

The **logical AND operator (&&)** is used to combine both conditions.
If both conditions are true, the program prints that the number is within the range. Otherwise, it displays a different message.

Task 3: If Statement

Program

```
#include <iostream>             // tells the compiler to include standard
input output library
using namespace std;             // allows us to use cout and cin without
having the need to type std:: for each

int main() {                // execution of main function beginns from here
      int a;                  // variable declaration

      cout<<"enter an integer ";     // prompt for user to input data
      cin>>a;                     // takes data from user and puts it in
variable

      if(a>0)                              // condition if a is grater than 0
      cout<<"the number is positive"<<endl;  // prints message if a is
greater than 0
      else                                 // runs if the if condition
fails
      cout<<"the number is negative"<<endl;  // prints the message if the
number is smaller than 0

      return 0;               // tells the operating system that program has
finishe

}
```

Explanation

In this program, the **if statement** checks whether the entered number is **greater than zero**.

If the condition is true, the program prints that the number is positive.
If the condition is false, the program simply ends without printing anything.

Task 4: If-Else Statement

Program

```
#include <iostream>  // tells the compiler to include standard input output
stream library
using namespace std;  // allows us to use cout and cin without having the
need to typ std:: for each cin and cout

int main() {                 // main function execution beggins from here

      int a;                                // variable declarartion
      cout<<"enter a number";             // displays are request to the user
      cin>>a;                                     // stores the user input
data into variable
      if(a>0)                                  // cheaks if the number is
greater than 0
      cout<<"the number is positive";             // displays thre message
if the number is greater than 0
      else if(a<0)                            // if test 1 fails , it
cheaks if the number is less than o
      cout<<"the number is negative";             // confirms that the
number is negative
      else if(a==0)                           // cheaks that if the number
is exactly 0
      cout<<"the number you entered is zero";         // final confirmation
for 0

      return 0;                            // tells the operating system that
program has finishe


}
```

Explanation

This program asks the user to enter their age.

The **if-else statement** checks whether the age is **greater than or equal to 18**.

- If the condition is true, the program prints that the person is eligible.
- If the condition is false, it prints that the person is not eligible.

Task 5: Else-If Statement

Program

```
#include <iostream>  // tells the compiler toinclude input output stream
library
using namespace std; // uses standard namespaces without having to write
std:: for every cin and cout

int main() {             //the main function from where the execution starts

      int age;                        // variable declaration
      cout<<"enter your age";       // asks the user to provide their age
      cin>>age;                    // takes the data and puts i into the
variable
      if(age>=18)                 //cheaks if the value in age is 18 or
greater
      cout<<"you are an edult"<<endl;       // if the condition mets it
prints you are an adult now
      else                                  // this part rund if the if
condition fails
      cout<<"you are a minor"<<endl;          // prints you are a minor

      return 0;                        // tells the operating system that
program has finishe

}
```

Explanation

This program determines whether a number is **positive, negative, or zero**.

- If the number is greater than zero, it is positive.
- If the number is less than zero, it is negative.
- If neither condition is true, the number must be zero.

The **else-if statement** allows checking multiple conditions sequentially.